Year 11 Mathematics

Algebra 7

2.3 Further Formulae

Further Formulae

STARTER · 5 MIN

Date: _____ **Title:** Further Formulae

LEARNING OBJECTIVES

- Rearrange formulae where the subject appears twice
- Factorise to change the subject
- Rearrange formulae with the subject in a fraction

STARTER *Answer in your book. No calculators.*

1. Factorise $3x + 6$
2. Factorise $ax + bx$
3. Expand $y(x - 2)$
4. Make x the subject of $y = 4x - 3$

Starter — Answers

STARTER

Tick or correct in green pen

1. $3(x + 2)$

2. $x(a + b)$

3. $xy - 2y$

4. $x = \frac{y + 3}{4}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can rearrange formulae where the subject appears twice	☐
SILVER	I can factorise to change the subject	☐
GOLD	I can rearrange formulae with the subject in a fraction	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Subject Appears Twice

INTRODUCTION · 3 MIN

Collect the subject terms on one side, then factorise.

Step 1

Expand any brackets.

$y(x + 2) = xy + 2y$

Step 2

Put the subject terms on one side.

$ax - bx = c$

Step 3

Factorise out the subject.

$x(a - b) = c$

Step 4

Divide by the bracket.

$$x = \frac{c}{a - b}$$

Examples 1 and 2 — Collect and Factorise

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Make x the subject of

$ax + b = cx$

Example 2

Make x the subject of

$3x + y = px$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$b = cx - ax$$

$$b = x(c - a)$$

$$x = \frac{b}{c - a}$$

Tip: get both x terms on one side.

Example 2

$$y = px - 3x$$

$$y = x(p - 3)$$

$$x = \frac{y}{p - 3}$$

Tip: factorise x out.

Task 1

PRACTICE · 9 MIN

BRONZE *Make x the subject*

1. $ax = bx + c$
2. $5x + k = mx$
3. $px - q = rx$
4. $x + y = ax$
5. $ax + bx = 7$

★ $tx - 4 = 2x + y$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $x = \frac{c}{a - b}$

2 $x = \frac{k}{m - 5}$

3 $x = \frac{q}{p - r}$

4 $x = \frac{y}{a - 1}$

5 $x = \frac{7}{a + b}$

★ $x = \frac{y + 4}{t - 2}$

Examples 3 and 4 — Brackets

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Make x the subject of

$y(x + 2) = 3x$

Example 4

Make x the subject of

$a(x - 1) = b(x + 1)$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$xy + 2y = 3x$$

$$2y = 3x - xy$$

$$2y = x(3 - y)$$

$$x = \frac{2y}{3 - y}$$

Tip: expand first.

Example 4

$$ax - a = bx + b$$

$$ax - bx = a + b$$

$$x = \frac{a + b}{a - b}$$

Tip: collect x terms on the left.

Task 2

PRACTICE · 9 MIN

SILVER

Make x the subject

1. $y(x - 1) = 2x$
2. $k(x + 3) = 5x$
3. $p(x + 2) = q(x - 1)$
4. $3(x + y) = a(x - y)$
5. $m(2x - 1) = x + n$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $x = \frac{y}{y - 2}$

2 $x = \frac{3k}{5 - k}$

3 $x = \frac{-q - 2p}{p - q}$

4 $x = \frac{-3y - ay}{3 - a}$

5 $x = \frac{m + n}{2m - 1}$

Examples 5 and 6 — Subject in a Fraction

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5 (3 marks)

Make x the subject of

$$y = \frac{x + 1}{x - 2}$$

Example 6 (3 marks)

Make a the subject of

$$b = \frac{3a}{a + c}$$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$y(x - 2) = x + 1$$
$$xy - 2y = x + 1$$
$$xy - x = 2y + 1$$
$$x = \frac{2y + 1}{y - 1}$$

Tip: multiply by the denominator first.

Example 6

$$ab + bc = 3a$$
$$bc = 3a - ab$$
$$bc = a(3 - b)$$
$$a = \frac{bc}{3 - b}$$

Tip: collect a terms, then factorise.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. Make x the subject: $y = \frac{x}{x+4}$ (3 marks)

2. Make x the subject: $y = \frac{2x-1}{x+3}$ (3 marks)

3. Make p the subject: $q = \frac{p+5}{p}$ (3 marks)

4. Make x the subject: $a = \frac{x+b}{x-b}$ (3 marks)

5. Make r the subject: $A = \frac{r+1}{3-r}$ (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $x = \frac{4y}{1 - y}$

2 $x = \frac{3y + 1}{2 - y}$

3 $p = \frac{5}{q - 1}$

4 $x = \frac{ab + b}{a - 1}$

5 $r = \frac{3A - 1}{A + 1}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Make x the subject of $ax = 3x + 2$

SILVER Make x the subject of $y(x + 1) = 4x$

GOLD Make x the subject of $y = \frac{x}{x - 1}$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x = \frac{2}{a - 3}$

SILVER $x = \frac{y}{4 - y}$

GOLD $x = \frac{y}{y - 1}$

Exam Practice — Collect and Factorise

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1 (3 marks)

Make x the subject of $ax + 3 = bx + 7$

.....

Question 2 (3 marks)

Make p the subject of

$5p - q = rp + 2$

.....

Exam Practice — Brackets

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3 (3 marks)

Make x the subject of

$y(x + 3) = 2x - 1$

.....

Question 4 (3 marks)

Make x the subject of

$a(x - 2) = b(x + 5)$

.....

Exam Practice — Subject in a Fraction

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(3 marks)

Make x the subject of

$$y = \frac{x + 2}{x - 3}$$

.....

Question 6

(4 marks)

Make a the subject of

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$$

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$ax - bx = 7 - 3$ — M1

$x(a - b) = 4$ — M1

$\mathbf{x = \frac{4}{a - b}}$ — A1

Tip: x terms on one side, the rest on the other.

Question 2

$5p - rp = q + 2$ — M1

$p(5 - r) = q + 2$ — M1

$\mathbf{p = \frac{q + 2}{5 - r}}$ — A1

Tip: factorise p out.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$xy + 3y = 2x - 1$ — M1

$xy - 2x = -1 - 3y$ — M1

$\mathbf{x = \dfrac{-1 - 3y}{y - 2}}$ — A1

Tip: collect the x terms on the left.

Question 4

$ax - 2a = bx + 5b$ — M1

$ax - bx = 5b + 2a$ — M1

$\mathbf{x = \dfrac{2a + 5b}{a - b}}$ — A1

Tip: expand both brackets first.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$y(x - 3) = x + 2$ — M1

$xy - x = 2 + 3y$ — M1

$\mathbf{x = \dfrac{3y + 2}{y - 1}}$ — A1

Tip: multiply by the denominator first.

Question 6

$\times abc$: $bc + ac = ab$ — M1

$bc = ab - ac$ — M1

$bc = a(b - c)$ — M1

$\mathbf{a = \dfrac{bc}{b - c}}$ — A1

Tip: clear all the fractions at once.